

A unified V_{600} scaffolding for finite-group derivations of 1/4 incidence, monochromatic σ -pair energy, Galois-coset duality, and operator-trace ratios

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Status: Preprint (not peer-reviewed). This is a SYNTHESIS paper. It cites four prior papers in the V_{600} programme and adds NO new theorem-grade physical claim. Its sole contribution is the explicit identification of a shared structural tuple $\Sigma_{V_{600}}$ that all four imported theorems factor through.

Abstract

We synthesise four prior results on the binary icosahedral group $V_{600} = 2I$: the Bekenstein–Hawking 1/4 coefficient as Dic₅-coset incidence on V_{600} [Smart, 2026a]; the discrete Hawking quantum $E_q = 5/2$ from σ -pair excitations on V_{600} [Smart, 2026b]; the canonical involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ with explicit phase-0 lift and $\mathbb{Z}_2^3 = 32$ phase ambiguity [Smart, 2026c]; and the exact rational operator-trace ratios $\text{tr}(I_{C_Q} + P_{72})/12 = 13/12$ and $\text{tr}(I_{C_Q} - P_0)/12 = 11/12$ on the 12-dimensional rational T_τ -cycle space, with the cosmological coupling rule stated as an explicit Layer-3 hypothesis [Smart, 2026d]. We record the load-bearing structural tuple

$$\Sigma_{V_{600}} = (2I, T_\tau\text{-cycles, K-multiset } \{72:1, 0:1, 52:5, 20:5\}, \text{Dic}_5, V_{24}, \text{right cosets Dic}_5g, \sigma, \tau_\sigma, \mathcal{A}_K),$$

and show that all four theorem-grade imports, with Paper 4 restricted to its Layer 1 trace identities and admissibility theorem, factor through it. The synthesis adds NO new theorem-grade physical claim. We do NOT claim a unified physical theory; we report a unified mathematical scaffolding. CMB-bulk projection, cosmic dipole ladder audits, and pre-registered prediction manifests are explicitly named future builds with unmet prerequisites; they are out of scope here.

1 Introduction

The V_{600} programme is five papers. Papers 1–4 each prove a single narrow finite-group identity:

- Paper 1 [Smart, 2026a]: per-coset incidence count $|C \cap V_{24}| = 4$ for $C \in \{g\text{Dic}_5, \text{Dic}_5g\}$, hence $S/A = 4/16 = 1/4$.
- Paper 2 [Smart, 2026b]: $|v - \sigma(v)|_E^2 = 5/2$ for every σ -mobile vertex of V_{600} , hence a single-line σ -pair excitation spectrum.
- Paper 3 [Smart, 2026c]: explicit τ_σ involution fixing Dic₅ pointwise, swapping $K = 52$ and $K = 20$ cycle classes within each non-bulk right coset, and commuting with T_τ and the antipodal map.
- Paper 4 [Smart, 2026d]: $\text{tr}(I_{C_Q} + P_{72}) = 13$ and $\text{tr}(I_{C_Q} - P_0) = 11$ on the 12-dimensional cycle space, plus a K-saturated admissibility theorem and an explicit Layer-3 cosmological coupling hypothesis.

The present paper is the synthesis. It does NOT prove anything new about physical reality. It identifies the cross-cutting finite structure that makes all four narrow theorems work, and records that identification as one structural-spine table plus four citation-only imported-theorem boxes. Anything that would require new theorems about cosmological projection, observed CMB, or large-scale-survey predictions is named out-of-scope (Section 8).

2 Structural spine

We collect the load-bearing finite data into one tuple.

Definition 2.1 (Structural spine). *The structural spine of the V_{600} programme is the finite tuple*

$$\Sigma_{V_{600}} = (2I, T_\tau, K\text{-multiset}, \text{Dic}_5, V_{24}, \mathcal{R}, \sigma, \tau_\sigma, \mathcal{A}_K),$$

whose components are the load-bearing facts of Table 1. All entries are imported from the cited foundation papers; no entry is introduced or re-proved here.

2.1 Correctness lemma: $V_{24} \neq \text{Dic}_5$

A subtle conflation in earlier exploratory drafts must be flagged.

Lemma 2.1 (V_{24} / Dic_5 distinction; Paper 1). *$\text{Fix}_\sigma(V_{600}) = V_{24}$ has 24 vertices (8 axis + 16 half-type). Dic_5 is the bulk subgroup of order 20 formed by the union of the $K = 72$ and $K = 0$ cycles. These are distinct subsets of V_{600} ; $|V_{24}| = 24$, $|\text{Dic}_5| = 20$, and $|\text{Dic}_5 \cap V_{24}| = 4$. More generally, $|C \cap V_{24}| = 4$ for every left or right coset $C \in \{g\text{Dic}_5, \text{Dic}_5g\}$ — this is the Bekenstein incidence theorem (Paper 1, not a new claim of this paper).*

Proof. Imported from Paper 1 [Smart, 2026a]. The 4/16 ratio underlies the Bekenstein 1/4. $\square$

2.2 Phase-independence

Lemma 2.2 (Cross-cutting phase-independence; Papers 3, 4). *The theorem-grade identities and properties imported below (Sections 3–6) are independent of which canonical τ_σ lift (out of the $\mathbb{Z}_2^5 = 32$ canonical lifts) is chosen. Specifically: the K -class projector traces, the $|gH \cap V_{24}|$ count, the σ -pair-energy spectrum, and the involution (P1)–(P4) properties are all functions of $\Sigma_{V_{600}}$ that do not depend on the phase choice. (The explicit phase-0 lift exhibited in Paper 3 is, by construction, phase-specific as a 120-vertex map; the structural properties it satisfies are phase-independent.)*

Proof. Phase-independence of (P1)–(P4) is in Paper 3 [Smart, 2026c] (all 32 lifts satisfy the same properties). Phase-independence of the operator-trace ratios is verified explicitly in Paper 4 [Smart, 2026d] (all 32 lifts yield cycle-level permutations that pointwise fix $\hat{H}$ and $\hat{C}$). The 1/4 incidence count and the σ -pair spectrum are properties of V_{600} and σ alone and do not reference τ_σ . $\square$

3 Imported theorem 1: Bekenstein 1/4 incidence (Paper 1)

Imported theorem (citation only) (Smart, 2026a). For every coset $C \in \{g\text{Dic}_5, \text{Dic}_5g\}$ of the bulk subgroup $\text{Dic}_5 \subset 2I$, the intersection $|C \cap V_{24}| = 4$, equivalently 4 σ -fixed vertices and 16 σ -mobile vertices per coset (left or right). The structural ratio $S/A = 4/16 = 1/4$ matches the Bekenstein–Hawking entropy coefficient.

Role in synthesis. Imported theorem 1 uses $(V_{600}, \text{Dic}_5, V_{24})$ from $\Sigma_{V_{600}}$. The result is purely arithmetic and does not require τ_σ or $\mathcal{A}_K$. Paper 1 is the load-bearing setup paper for the rest of the programme; its right-coset/left-coset distinction (Lemma in Paper 1) is the load-bearing fact for Papers 3 and 4.

4 Imported theorem 2: Hawking quantum $E_q = 5/2$ (Paper 2)

Imported theorem (citation only) (Smart, 2026b). Every σ -mobile vertex $v \in V_{600} \setminus V_{24}$ satisfies $|v - \sigma(v)|_E^2 = 5/2$ in the Euclidean norm. The 96 σ -mobile vertices yield a monochromatic σ -pair excitation spectrum $24 \cdot \delta_0 + 96 \cdot \delta_{5/2}$. A finite first-law identity holds per coset (left or right; the bulk and non-bulk cases share the same per-coset arithmetic): $\mathcal{T}_{\text{casc}} \cdot S = E/4$ with $A = 16$, $S = 4$, $E = 40$, $\mathcal{T}_{\text{casc}} = 5/2$, where $\mathcal{T}_{\text{casc}}$ is the finite cascade temperature parameter of Paper 2 (NOT calibrated to a physical Hawking temperature; see Paper 2’s explicit caveat).

Role in synthesis. Imported theorem 2 uses $(V_{600}, V_{24}, \sigma, \text{Dic}_5)$ from $\Sigma_{V_{600}}$. It is independent of T_τ , τ_σ , and $\mathcal{A}_K$. The first-law identity reuses the 4/16 incidence from Paper 1.

5 Imported theorem 3: τ_σ involution (Paper 3)

Imported theorem (citation only) (Smart, 2026c). There exists an explicit involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ such that (P1) $\tau_\sigma \circ \tau_\sigma = \text{id}$; (P2) τ_σ fixes Dic_5 pointwise; (P3) τ_σ swaps the $K = 52$ and $K = 20$ cycle classes within each non-bulk right coset of Dic_5 ; (P4) τ_σ commutes with T_τ (i.e., $\tau_\sigma \circ T_\tau = T_\tau \circ \tau_\sigma$); and additionally τ_σ commutes with the antipodal map $v \mapsto -v$. There is a $\mathbb{Z}_2^5 = 32$ phase ambiguity in the canonical lift; the phase-(0,0,0,0,0) representative is explicit on 120 vertices.

Role in synthesis. Imported theorem 3 uses $(V_{600}, T_\tau, K\text{-multiset}, \text{Dic}_5, \mathcal{R}, \sigma)$ from $\Sigma_{V_{600}}$ and produces τ_σ . The construction is via right cosets $\mathcal{R}$ and the icosian trace metric; the phase-independence Lemma 2.2 is the load-bearing fact downstream.

6 Imported theorem 4 (Paper 4): three layers

Paper 4 maintains a strict three-layer credibility split. We import *only Layer 1* as a theorem (IT4); Layer 2 and Layer 3 are brought across separately as numerical observation and explicit hypothesis, neither of which is promoted by this synthesis.

Imported theorem (citation only) (Smart, 2026d, Layer 1 only). On the 12-dimensional rational cycle space $C_{\mathbb{Q}} \cong \mathbb{Q}^{12}$:

1. *Trace identities.* $\text{tr}(I_{C_{\mathbb{Q}}} + P_{72}) = 13$ and $\text{tr}(I_{C_{\mathbb{Q}}} - P_0) = 11$, hence ratios 13/12 and 11/12 as exact rationals.
2. *K-saturated admissibility.* Within $\mathcal{A}_K$, the only rank-one projector corrections to $I_{C_{\mathbb{Q}}}$ are $\pm P_{72}$ and $\pm P_0$.

Layer 2 (observation). Paper 4 reports $\frac{13}{12} \cdot H_0^{\text{Planck}} \approx 72.97 \text{ km/s/Mpc}$ versus SH0ES 73.04 ± 1.04 ($\approx 0.06\sigma$ residual), and $\frac{11}{12} \cdot S_8^{\text{Planck}} \approx 0.763$ versus KiDS-1000 multi-probe $0.766^{+0.020}_{-0.014}$ ($\approx 0.24\sigma$ directional residual using the lower KiDS error). These are observational, not derivations.

Layer 3 (hypothesis). Paper 4 states the cycle-mode coupling hypothesis Γ explicitly (Smart, 2026d, H1). The synthesis carries this forward as the same hypothesis; we do NOT promote it to a theorem here.

Role in synthesis. Imported theorem 4 uses $(V_{600}, T_\tau, \text{K-multiset}, \mathcal{A}_K)$ from $\Sigma_{V_{600}}$. Phase-independence (Lemma 2.2) ensures the trace ratios do not depend on the choice of canonical τ_σ lift.

7 Synthesis

Proposition 7.1 (Common factorisation through $\Sigma_{V_{600}}$). *Each of imported theorems 1, 2, 3, and IT4-Layer 1 is a statement about $\Sigma_{V_{600}}$. Specifically:*

- *Theorem 1 (Bekenstein 1/4): a function of $(V_{600}, \text{Dic}_5, V_{24})$.*
- *Theorem 2 (Hawking $E_q = 5/2$): a function of $(V_{600}, V_{24}, \sigma, \text{Dic}_5)$ with the inherited Euclidean icosian norm on the underlying $\mathbb{H}_K$.*
- *Theorem 3 (τ_σ structural-property package (P1)–(P4) + antipodal compatibility): a function of $(V_{600}, T_\tau, \text{K-multiset}, \text{Dic}_5, \mathcal{R}, \sigma, \tau_\sigma)$. Note: the explicit phase-0 vertex map of Paper 3 also requires the icosian trace metric and the deterministic base-point/ordering conventions of Paper 3’s `vfd_v600.icosian.build_vertices()` as auxiliary data; those conventions are not part of $\Sigma_{V_{600}}$ but are not needed for the structural-property package itself.*
- *IT4-Layer 1 (trace ratios + K-saturated admissibility): a function of $(V_{600}, T_\tau, \text{K-multiset}, \mathcal{A}_K)$.*

None of the four imported Layer 1 theorems uses any datum outside $\Sigma_{V_{600}}$. The factorisation does NOT extend to IT4 Layer 2 (observational benchmarks H_0 , S_8 from external cosmology data) or IT4 Layer 3 (the coupling hypothesis Γ relating the trace ratios to those observables); both Layer 2 and Layer 3 explicitly invoke data and physical hypotheses outside $\Sigma_{V_{600}}$.

Proof. By inspection of each cited paper. The full derivations are in Smart [2026a,b,c,d]; this paper re-proves nothing. $\square$

Remark 7.1 (What the synthesis does and does not say). *Proposition 7.1 is a structural observation: four finite-group results share a common load-bearing structure. It is NOT a unification theorem in physics; it does not derive black-hole thermodynamics, Hawking radiation, or cosmological tensions. It records that all four identities depend on the same underlying tuple. Whether this shared dependence has physical content is a separate question that this paper does not address.*

8 Honest scope and named future builds

The credibility-bar discipline of Papers 1–4 is preserved. We explicitly name three classes of work that would extend the synthesis but are NOT claims of this paper:

(F1) CMB-bulk projection bridge. The closure-cosmogenesis exploratory note in this repository (`docs/closure-cosmogenesis.md`) contains conditional residual and coarse-graining ingredients — a conditional bulk-invariance property (residual support $\rho_t^B = 0$ is conditional), an unspecified coarse-graining π_C with open weight choices, and no temperature calibration — that a future bridge map $\Theta : (F_t^B, \pi_C, \text{calibration}) \rightarrow T : S^2 \rightarrow \mathbb{R}_+$ from a candidate bulk-supported sector of states (with σ -invariance to be proved by the bridge, not assumed) to an observed CMB temperature field would have to complete. The source note does NOT contain such a map. The bulk-supported sector here is NOT identified with V_{24} : the bulk subgroup Dic_5 has 20 elements and V_{24} has 24; the two are distinct (Lemma 2.1), with $|\text{Dic}_5 \cap V_{24}| = 4$. We do NOT use any of this in the synthesis; “uniform T_{CMB} follows from bulk invariance” is NOT a theorem of this paper, and any source language identifying the 20-vertex bulk with the 24-vertex σ -fixed sublattice is a known conflation that this synthesis explicitly disowns.

(F2) Cosmic dipole / $1 + k/2$ ladder audit. Repository scripts catalogue a ladder of cascade-resolution predictions and report agreement with several survey dipole-axis measurements. Those scripts are exploratory and partly fitted; the k -rule in `papers/cosmological-folding-rate/dyna` is a working hypothesis, not a frozen pre-registration. We do NOT claim any dipole result in the synthesis; a separate empirical audit with frozen survey inclusion, uncertainties, k -assignment, and null model is needed before any such claim is defensible.

(F3) Pre-registered prediction manifest. A locked, timestamped prediction manifest with formula, value, uncertainty window, and source commit per observable would convert Hypothesis Γ (Paper 4) into a falsifiable programme. The present scripts (`preregister_k.py`, `cmb_mapping.py`, the legacy `UNIFIED_RESULT.md`) are not in that form. The synthesis does NOT claim a prediction catalogue; we cite the existing exploratory artifacts only as programme context.

What the synthesis claims, in one sentence. There is a single finite tuple $\Sigma_{V_{600}}$ such that the four imported theorems factor through it. That is the entire claim.

9 Conclusion

The four foundation papers of the V_{600} programme each prove a narrow finite-group identity. This synthesis records that they all factor through one structural tuple $\Sigma_{V_{600}}$. No new theorem-grade physical claim is added; CMB, dipole, and prediction-manifest work is named out-of-scope with explicit unmet prerequisites.

A Verification certificate

The companion script `verify.py` computes $\Sigma_{V_{600}}$ from `vfd_v600.group.build_state()` (using only the Python standard library and the shared `vfd_v600` package), and asserts the listed headline cross-checks:

1. $|V_{600}| = 120$, 12 T_7 -cycles each of length 10, K-multiset $\{72:1, 0:1, 52:5, 20:5\}$.
2. $|\text{Dic}_5| = 20$ and $|V_{24}| = 24$, so $\text{Dic}_5 \neq V_{24}$ (Lemma 2.1).
3. Bekenstein cross-check: $|C \cap V_{24}| = 4$ for every left or right coset $C \in \{g\text{Dic}_5, \text{Dic}_5g\}$; ratio $4/16 = 1/4$.
4. Hawking cross-check: $|v - \sigma(v)|_E^2 = 5/2$ for every σ -mobile vertex; spectrum is $24 \cdot \delta_0 + 96 \cdot \delta_{5/2}$.
5. τ_σ cross-check (canonical phase-0 lift): (P1)–(P4) properties + antipodal compatibility hold.
6. Trace-ratio cross-check: $\text{tr}(I_{C_{\mathbb{Q}}} + P_{72}) = 13$ and $\text{tr}(I_{C_{\mathbb{Q}}} - P_0) = 11$ on $C_{\mathbb{Q}} \cong \mathbb{Q}^{12}$.
7. Phase-independence: for each of the $2^5 = 32$ canonical τ_σ phase lifts, the induced cycle-level action pointwise fixes both $I_{C_{\mathbb{Q}}} + P_{72}$ and $I_{C_{\mathbb{Q}}} - P_0$.

Reproducibility command:

```
PYTHONPATH=papers/v600-programme/lib \
python3 papers/v600-unified/verify.py
```

References

- John H. Conway and Neil J. A. Sloane. *Sphere Packings, Lattices and Groups*. Springer-Verlag, 3rd edition, 1988.
- H. S. M. Coxeter. *Regular Polytopes*. Dover, 3rd edition, 1973.
- Lee Smart. A $1/4$ incidence identity on the 600-cell and its Bekenstein–Hawking analogy. V_{600} programme, Paper 1., 2026a.
- Lee Smart. A discrete Hawking quantum from σ -pair excitations on v_{600} . V_{600} programme, Paper 2., 2026b.
- Lee Smart. The τ_σ involution: A canonical Galois-coset duality on the binary icosahedral group. V_{600} programme, Paper 3., 2026c.
- Lee Smart. Two operator-trace ratios from v_{600} k-class structure: $13/12$ and $11/12$, and their match to the h_0 and s_8 benchmarks. V_{600} programme, Paper 4., 2026d.

Table 1: Components of the structural spine $\Sigma_{V_{600}}$. Every entry is imported from the cited paper or from the shared library `vfd_v600`; this paper does not re-prove any of them.

Component	Description	Source	Status
$2I = V_{600}$	Binary icosahedral group; 120 unit icosian quaternions; vertices of the regular 600-cell.	Coxeter [1973] , Conway and Sloane [1988] , Paper 1 setup	exact
T_τ -cycles	$T_\tau : g \mapsto \tau g$ partitions V_{600} into 12 cycles each of length 10, for any $\tau \in 2I$ of order 10.	Paper 1	exact
K-multiset	The 12 cycles carry K-class assignments $\{72:1, 0:1, 52:5, 20:5\}$.	Paper 1	exact
Dic_5	Binary dihedral subgroup of order 20; equals the union of the $K = 72$ and $K = 0$ cycles; self-normalising and non-normal in $2I$.	Paper 1	exact
V_{24}	σ -fixed sublattice $\text{Fix}_\sigma(V_{600})$; 24 vertices; the 24-cell (8 axis + 16 half-type).	Paper 1	exact
σ	The $K/\mathbb{Q}$ Galois twist $\sqrt{5} \mapsto -\sqrt{5}$ on the icosian quaternion algebra $\mathbb{H}_K$, restricted to act on V_{600} . Defines $V_{24} = \text{Fix}_\sigma$ and the σ -pair energy $ v - \sigma(v) _E^2$.	Paper 1, Paper 2	exact
$\mathcal{R}$	The five non-bulk right cosets $\text{Dic}_5 g$, each the disjoint union of one whole $K=52$ cycle and one whole $K=20$ cycle. Left cosets $g\text{Dic}_5$ are NOT whole-cycle carriers.	Paper 1, Paper 3	exact
τ_σ	Canonical involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ with phase-0 lift explicit on 120 vertices; $\mathbb{Z}_2^5 = 32$ phase ambiguity; fixes Dic_5 pointwise; swaps $K = 52 \leftrightarrow K = 20$ within each non-bulk right coset; commutes with T_τ and the antipodal map.	Paper 3	exact
$\mathcal{A}_K$	K-saturated diagonal projector algebra $\langle I_{C_\mathbb{Q}}, P_{72}, P_0, P_{52}, P_{20} \rangle_\mathbb{Q}$ on the 12-dimensional rational cycle space $C_\mathbb{Q} \cong \mathbb{Q}^{12}$.	Paper 4	exact